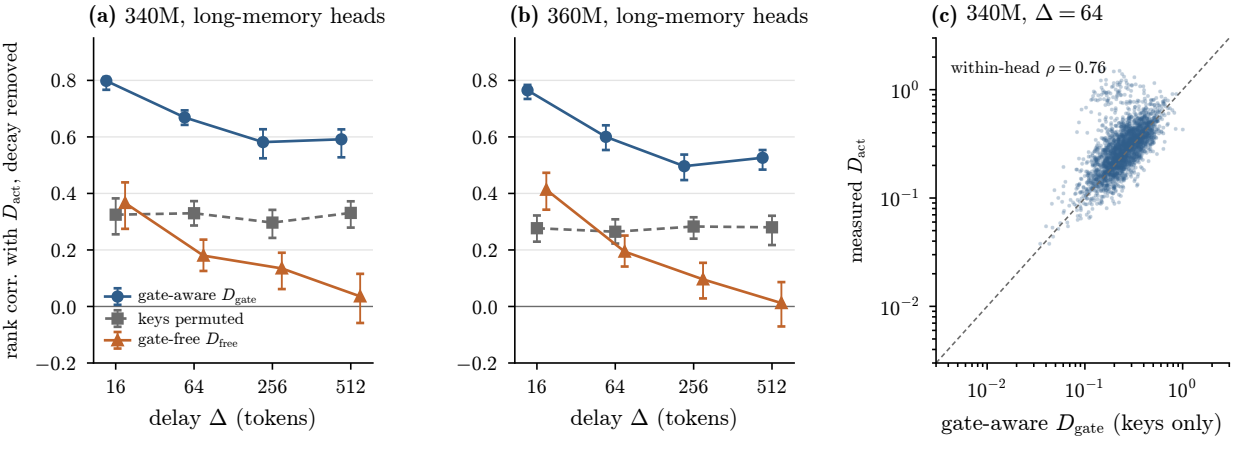
(a) 340M, long-memory heads
(b) 360M, long-memory heads
(c) 340M, Δ = 64
rank corr. with D_{act}, decay removed
0.8
0.6
0.4
0.2
0.0
−0.2
16
64
256
512
delay Δ (tokens)
gate-aware D_{gate}
keys permuted
gate-free D_{free}
within-head $\rho = 0.76$
measured D_{act}
10^{0}
10^{-1}
10^{-2}
gate-aware D_{gate} (keys only)